
Workshop 1: Multi-Armed Bandits

Exploration, Regret and Algorithm Comparison

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Experimental setup

10-armed bandits

Number of arms: $k = 10$,

Expected reward: $q_*(a) \sim \mathcal{N}(0, 1)$,

Randomised reward: $R_t \mid A_t = a \sim \mathcal{N}(q_*(a), 1)$.

Fixed testbed, seed 5. The optimal arm is $a_* = 2$ in Python indexing (the 3rd arm), with $q_*(a_*) = 2.4308$.

Simulation settings

Tasks	Horizon, T	Rerun, M
Q3	10^5	500
Q4–Q7	10^4	300

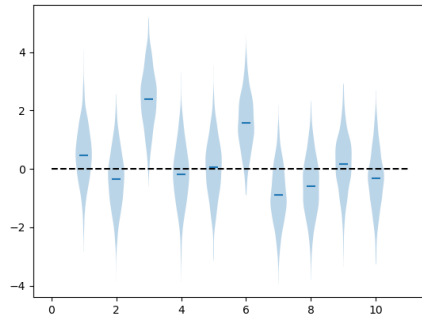


Figure 1: Reward distributions and true action values $q_*(a)$ for the fixed ten-arm bandit. The same fixed q_* is used throughout.

What does sublinear regret mean?

$$\text{Regret}_T = \mathbb{E} \left[\sum_{t=0}^{T-1} \left(q_*(a_*) - q_*(A_t) \right) \right],$$
$$\text{Regret}_T = o(T) \iff \lim_{T \rightarrow \infty} \frac{\text{Regret}_T}{T} = 0.$$

Sublinear regret means that cumulative regret grows slower than linearly with the horizon T , so the average regret per time step converges to zero.

Simulation estimator

$$\hat{R}_T = \frac{1}{M} \sum_{r=1}^M \sum_{t=0}^{T-1} \left[q_*(a_*) - q_*(A_t^{(r)}) \right].$$

Why it matters

- The long-run average reward approaches the optimal expected reward $q_*(a_*)$.
- For every strictly suboptimal arm, the fraction of time it is selected vanishes as $T \rightarrow \infty$.
- Exploration may continue, provided its cumulative regret grows sublinearly with the horizon.

How do we test sublinearity numerically?

1. Normalised regret

$$\hat{\rho}_T = \frac{\hat{R}_T}{T}$$

Sublinear behaviour requires $\hat{\rho}_T$ to trend toward zero as the horizon grows.

2. Incremental slope

$$\hat{s}_i = \frac{\hat{R}_{T_{i+1}} - \hat{R}_{T_i}}{T_{i+1} - T_i}$$

A falling late-horizon slope indicates decreasing marginal regret.

3. Effective exponent

$$\hat{\alpha}_i = \frac{\log \hat{R}_{T_{i+1}} - \log \hat{R}_{T_i}}{\log T_{i+1} - \log T_i}$$

Values below one are consistent with $\hat{R}_T \propto T^\alpha$ over that interval.

Evaluation protocol

- Horizon $T = 10^5$.
- $M = 500$ independent reruns.
- Same bandit instance for every policy.
- Evaluate all three diagnostics jointly.

What does the effective exponent tell us?

Local power-law approximation

$$\hat{R}_T \approx CT^\alpha$$

$$\log \hat{R}_T \approx \log C + \alpha \log T$$

$$\Delta \log \hat{R}_T \approx \alpha \Delta \log T$$

$$\alpha \approx \frac{\Delta \log \hat{R}_T}{\Delta \log T}$$

$$\hat{\alpha}_i = \frac{\log \hat{R}_{T_{i+1}} - \log \hat{R}_{T_i}}{\log T_{i+1} - \log T_i}.$$

Thus, $\hat{\alpha}_i$ estimates the local slope of $\log \hat{R}_T$ against $\log T$.

Interpretation

$$\hat{\alpha}_i < 1$$

Regret grows slower than linearly over that interval; consistent with sublinear growth.

$$\hat{\alpha}_i \approx 1$$

Regret is approximately linear in T ; regret per step does not vanish.

$$\hat{\alpha}_i \rightarrow 1$$

A finite-horizon exponent below one may still trend toward linear growth.

ϵ -greedy method

ϵ -greedy experiment used in Q3

```

1:  for each tested schedule  $\epsilon_t$  do
2:      repeat independently for  $M = 500$  runs
3:      initialise  $Q(a) = 0$ ,  $N(a) = 0$  for all arms
4:      for  $t = 1, \dots, T$  do
5:           $A_t \leftarrow \begin{cases} \arg \max_a Q_t(a), & \text{with probability } 1 - \epsilon_t, \\ \text{uniformly random arm,} & \text{with probability } \epsilon_t \end{cases}$ 
6:          sample reward  $R_t \sim \mathcal{N}(q_*(A_t), 1)$ 
7:           $N(A_t) \leftarrow N(A_t) + 1$ 
8:           $Q(A_t) \leftarrow Q(A_t) + \frac{R_t - Q(A_t)}{N(A_t)}$ 
9:          accumulate pseudo-regret  $q_*(a_*) - q_*(A_t)$ 
10:     end for
11:     average cumulative regret over the  $M$  runs
12:     compute  $\hat{\rho}_{T_i}$ ,  $\hat{s}_i$ ,  $\hat{\alpha}_i$  across successive horizons
13: end for

```

Which schedules of ϵ -greedy show sublinear regret?

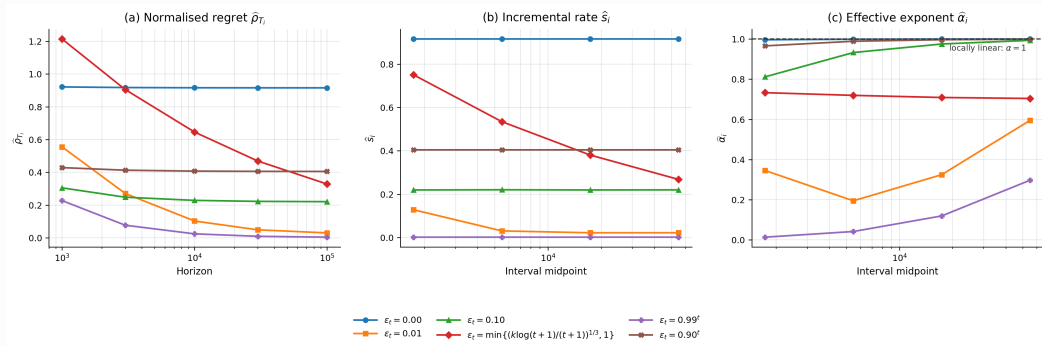


Figure 2: Three numerical diagnostics for the six ϵ -greedy schedules: (a) normalised regret $\hat{\rho}_{T_i}$, (b) incremental rate $\hat{s}_i$, and (c) effective exponent $\hat{\alpha}_i$.

Normalised regret measures regret per step, while the incremental slope and effective exponent describe how the regret growth rate changes with horizon.

Interpreting the exploration schedules

Schedule	Evidence	Diagnosis
$\varepsilon = 0$	$\hat{\rho}, \hat{s}$ constant; $\hat{\alpha} \approx 1$	Not sublinear
$\varepsilon = 0.01$	low rates; $\hat{\alpha}$ still increasing	Inconclusive
$\varepsilon = 0.10$	$\hat{\rho}, \hat{s}$ plateau; $\hat{\alpha} \approx 1$	Not sublinear
Polynomial	$\hat{\rho}, \hat{s} \downarrow, \hat{\alpha} < 1$	Sublinear
0.99^t	low rates; $\hat{\alpha}$ still increasing	Inconclusive
0.90^t	$\hat{\rho}, \hat{s}$ plateau; $\hat{\alpha} \approx 1$	Not sublinear

Fixed ε

Numerically, $\varepsilon = 0.01$ is still inconclusive; Theoretically, fixed $\varepsilon > 0$ samples inferior arms indefinitely, giving linear regret.

Polynomial decay

Exploration decreases gradually while remaining sufficient to correct earlier estimates.

Exponential decay

0.99^t is also numerically inconclusive; Rapidly vanishing exploration can cause premature commitment.

Polynomial decay gives the clearest evidence of sublinear regret.

UCB method

UCB experiment used in Q4–Q6

```

1:  for each  $c \in \{0, 1, 2\}$  do
2:    repeat independently for  $M = 300$  runs
3:    pull each arm once; initialise  $Q(a)$  and  $N(a)$ 
4:    for  $t = k + 1, \dots, T$  do

5:       $A_t \leftarrow \arg \max_a \left[ Q_t(a) + c \sqrt{\frac{\log(t+1)}{N_t(a)}} \right]$ 
6:      sample reward  $R_t \sim \mathcal{N}(q_*(A_t), 1)$ 
7:       $N(A_t) \leftarrow N(A_t) + 1$ 
8:       $Q(A_t) \leftarrow Q(A_t) + \frac{R_t - Q(A_t)}{N(A_t)}$ 
9:      accumulate pseudo-regret  $q_*(a_*) - q_*(A_t)$ 
10:    end for
11:    compute arm-wise error  $Q_T(a) - q_*(a)$ 
12:    average results across the  $M$  runs
13:  end for

```

Q4 uses the arm-value errors to compute RMSE; Q5–Q6 compare cumulative pseudo-regret.

UCB arm-value estimation

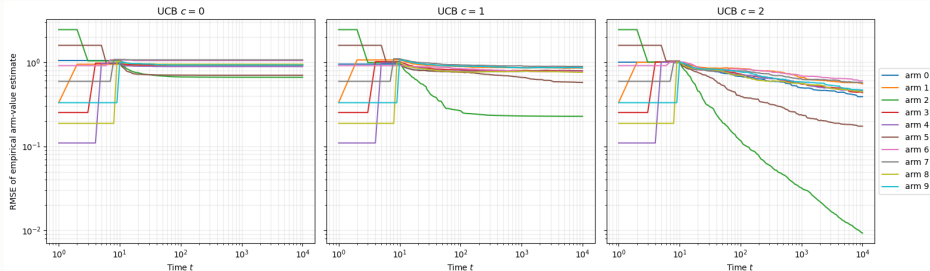


Figure 3: Evolution of the arm-value estimation error for UCB with $c = 0, 1, 2$.

Final mean arm-wise RMSE

UCB c	mean RMSE
0	0.8981
1	0.7254
2	0.4090

Per arm, $\text{RMSE}_T(a) = \sqrt{M^{-1} \sum_r (Q_T^{(r)}(a) - q_*(a))^2}$; the table averages this over ten arms.

Result: larger c estimates all arms better. This differs from minimising regret.

UCB versus epsilon-greedy regret

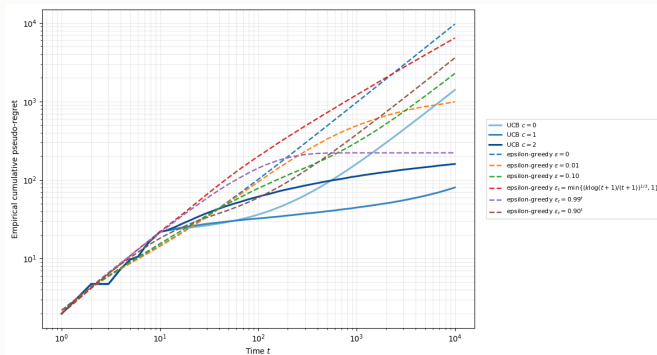


Figure 4: Cumulative regret comparison between UCB and selected ϵ -greedy policies.

Regret at $T = 10^4$

policy	$\hat{R}_T$
UCB $c = 1$	80.58
UCB $c = 2$	160.29
$\epsilon_t = 0.99^t$	221.98
$\epsilon = 0.01$	994.19

Lowest regret: $c = 1$.

Lowest RMSE: $c = 2$.

Index:

$$Q_t(a) + c\sqrt{\log(t+1)/N_t(a)}.$$

Gradient bandit method

Gradient-bandit experiment used in Q6–Q7

```

1:  for each tested learning rate  $\alpha$  do
2:    repeat independently for  $M = 300$  runs
3:    initialise  $H(a) = 0$  for all arms and  $\bar{R} = 0$ 
4:    for  $t = 1, \dots, T$  do
5:       $P_t(a) \leftarrow \frac{e^{H_t(a)}}{\sum_b e^{H_t(b)}}$ 
6:      sample  $A_t \sim P_t(\cdot)$  and observe  $R_t \sim \mathcal{N}(q_*(A_t), 1)$ 
7:       $H_{t+1}(a) \leftarrow H_t(a) + \alpha(R_t - \bar{R}_t)(1_{a=A_t} - P_t(a)), \quad \forall a$ 
8:       $\bar{R}_{t+1} \leftarrow \bar{R}_t + \frac{R_t - \bar{R}_t}{t}$ 
9:      accumulate pseudo-regret  $q_*(a_*) - q_*(A_t)$ ; record  $P_t(a_*)$ 
10:    end for
11:    average results across the  $M$  runs
12:  end for

```

Q6 compares cumulative pseudo-regret against UCB; Q7 examines whether $P_t(a_*) \rightarrow 1$.

Gradient bandit versus UCB

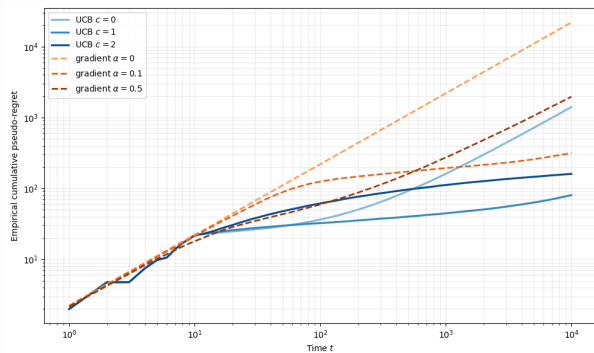


Figure 5: Cumulative regret comparison between gradient bandit and UCB policies.

Regret at $T = 10^4$

policy	$\hat{R}_T$
UCB $c = 1$	80.58
UCB $c = 2$	160.29
Grad. $\alpha = 0.1$	314.58
Grad. $\alpha = 0.5$	1968.16
UCB $c = 0$	1408.07

3.90× more regret

Best gradient versus UCB
 $c = 1$, on this experiment.

Gradient step-size sensitivity

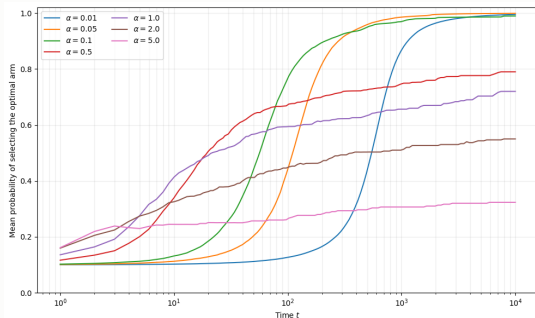


Figure 6: Effect of the gradient-bandit step size α on the probability of selecting the optimal arm.

Final performance at $T = 10^4$

α	mean $P_T(a_*)$	$\hat{R}_T$
0.01	0.9952	1542.45
0.05	0.9991	357.39
0.10	0.9896	314.58
0.50	0.7899	1968.16
2.00	0.5500	5358.04
5.00	0.3233	11148.71

At $\alpha = 5$, the final optimal-arm probability is nearly binary: 32.33% of runs have $P_T(a_*) > 0.95$, while 67.67% have $P_T(a_*) < 0.05$.

Large α causes unstable preference updates and premature lock-in; optimal-arm convergence is therefore not guaranteed for arbitrary $\alpha > 0$.

Conclusions

1. **Q1–Q2:** sublinear regret means vanishing regret per step; credible finite-horizon evidence needs several complementary diagnostics.
2. **Q3:** polynomially decaying exploration gives the clearest sublinear trend; a low average can conceal rare lock-in.
3. **Q4–Q5:** UCB $c = 2$ estimates arm values best, while UCB $c = 1$ achieves the lowest regret.
4. **Q6–Q7:** gradient bandit works well only in a moderate step-size range; aggressive updates can create bimodal outcomes.

All reported comparisons use the same fixed ten-arm Gaussian testbed and notebook protocols. Claims are empirical unless stated otherwise.

Rare-failure audit for $\varepsilon_t = 0.99^t$

Late-horizon regret rate

For each rerun r ,

$$s_{\text{late}}^{(r)} = \frac{R_{100000}^{(r)} - R_{30000}^{(r)}}{70000}.$$

499/500

$$s_{\text{late}}^{(r)} \approx 0$$

1/500

$$s_{\text{late}}^{(r)} = 0.84829$$

$R_T^{(r)}$: cumulative pseudo-regret of run r .

Mean late slope:

$$\frac{0.84829}{500} \approx 0.001697.$$

Interpretation

$$\sum_{t=0}^{\infty} 0.99^t = 100 < \infty$$

- Exploration eventually becomes negligible.
- 499/500 runs had already found the optimal arm.
- The failed run locked onto arm 5:

$$q^*(2) - q^*(5) \approx 0.84829,$$

exactly matching its late regret rate.

Takeaway: mean performance can hide a rare near-linear-regret failure.

UCB arm-level estimates at $T = 10,000$

arm	$q_*(a)$	mean $Q_T(a)$, $c = 2$	mean $N_T(a)$
0	0.441	0.316	9.3
1	-0.331	-0.459	5.3
2	2.431	2.430	9900.0
3	-0.252	-0.321	5.7
4	0.110	-0.004	7.2
5	1.582	1.526	44.0
6	-0.909	-1.058	3.8
7	-0.592	-0.755	4.4
8	0.188	0.090	7.6
9	-0.330	-0.423	5.3

Mean arm-wise RMSE

c	RMSE
0	0.8981
1	0.7254
2	0.4090

Even $c = 2$ allocates almost all pulls to the best arm. Its extra exploration primarily improves estimates of the remaining arms.

Full gradient step-size sweep

α	mean $P_T(a_*)$	regret	$P_T > 0.95$	$P_T < 0.05$
0.01	0.9952	1542.45	1.0000	0.0000
0.05	0.9991	357.39	1.0000	0.0000
0.10	0.9896	314.58	0.9900	0.0100
0.50	0.7899	1968.16	0.7900	0.2100
1.00	0.7200	2765.23	0.7200	0.2800
2.00	0.5500	5358.04	0.5500	0.4500
5.00	0.3233	11148.71	0.3233	0.6767

Why “between” is zero

The final-policy distribution is almost binary: each run ends either strongly on the optimal arm or strongly on another arm. Therefore

$$\Pr(0.05 \leq P_T(a_*) \leq 0.95) = 0$$

for every tested α . This column measures concentration, not statistical uncertainty.

Classic greedy and UCB algorithms

ϵ -greedy

- 1: initialise $Q(a) = 0$, $N(a) = 0$ for all arms a
- 2: **for** $t = 1, \dots, T$ **do**
- 3: with probability ϵ_t , choose random arm A_t
- 4: otherwise choose $A_t = \arg \max_a Q_t(a)$
- 5: observe reward R_t
- 6: update $N(A_t)$ and sample-average $Q(A_t)$
- 7: **end for**

UCB

- 1: initialise $Q(a) = 0$, $N(a) = 0$ for all arms a
- 2: pull each arm once
- 3: **for** $t = k + 1, \dots, T$ **do**
- 4: choose $A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\log t}{N_t(a)}} \right]$
- 5: observe reward R_t
- 6: update $N(A_t)$ and sample-average $Q(A_t)$
- 7: **end for**

Classic gradient bandit algorithm

```
1: initialise preferences  $H(a) = 0$  for all arms  $a$  and baseline  $\bar{R} = 0$ 
2: for  $t = 1, \dots, T$  do
3:   compute softmax policy  $P_t(a) = \frac{e^{H_t(a)}}{\sum_b e^{H_t(b)}}$ 
4:   sample action  $A_t$  according to  $P_t(a)$ 
5:   observe reward  $R_t$ 
6:   for every arm  $a$  do
7:     update  $H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t)(\mathbf{1}_{a=A_t} - P_t(a))$ 
8:   end for
9:   update baseline  $\bar{R}_{t+1} = \bar{R}_t + \frac{1}{t}(R_t - \bar{R}_t)$ 
10: end for
```

$H_t(a)$ is a preference, not a reward estimate. Positive reward surprise increases the selected arm's future probability.